Nick Vatanshenas

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Education

University of California, Los Angeles (UCLA) Expected Graduation: June 2028
B.S. Computer Science *4.0/4.0 GPA*

Irvine Valley College Graduated: May 2026
A.S. Computer Science *4.0/4.0 GPA*

Technical Skills

Languages: C/C++, C#, Python, Java, Kotlin, Assembly, Odin
Engines & Libraries: Unity, Godot, Raylib, OpenCV, Pandas, Tkinter
Developer Tools: Git, VS Code, CMake, Docker, Linux, Claude Code, Bitbucket, Jira

Projects — write-ups and source code on github

Custom Particle Engine | *C++, 3D Simulation* August 2026 – Present
- Building a **3D C++ particle engine** from scratch using object-oriented design, implementing Verlet integration to handle physical behavior and dynamic particle motion
- Implemented an **Octree spatial partitioning structure** in C++ to manage dynamic 3D spatial queries and organize particle spatial data

Holstein Collection Inc. (Godot Wild Jam #96) | *Lead Programmer, Godot, Python* August 2026
- **Empowered a narrative designer** on a **5-person team** by building a custom **Python** parser and Godot runtime loader, allowing them to write dialog and branching logic natively in Twine without touching game code
- Streamlined the creative workflow by translating **story constraints into a custom Godot dialogue system** for a visual novel game, securing a **top 12 finish out of 183 entries**

Trials of Yarbil | *Custom Engine, Odin (C-like language), Steam* January 2024 – August 2025
- Built and shipped a top-down action game on **Steam** using Odin and Raylib, writing a **custom game engine** to manage update order, custom state updates, and interactions across distinct game objects
- Developed engine subsystems (smoothed **grid-partitioned A*** pathfinding, level editors, game state serialization), and gameplay mechanics, while incorporating community feedback from playtests and devlogs (**1,000+ views**)

Professional Experience

Software Engineer Intern July 2026 – September 2026
XP Health
- Upgraded a **Python** LLM evaluation framework on a ∼**15-engineer team** using **Claude Code**, creating an eval dataset from session logs and stubbing MCP/API clients, backed by **50+ new unit tests** and developer docs
- Fixed a **Kotlin** race-condition with a **thread-safe state handler**, eliminating UI discrepancies for members
- Delivered production features for an **Android** vision store via design docs, code reviews, and **feature flag rollout**, impacting **2,300+ annual users** (**20% of active app base**) and completing core deliverables **1 week ahead of schedule**

Software Engineer Intern January 2026 – June 2026
Castix LLC
- **Profiled Unity performance bottlenecks** for an indie MMORPG, presenting a root-cause analysis on a ∼**25ms frame overhead**—caused by scene re-renders in a water plugin—**to the director** to guide optimization strategy
- Built **semi-automated Unity C# tools** to generate **primitive colliders from skeleton data** and automate prefab spawning, achieving a **10x setup speedup** for 400+ creatures and documenting handoffs for lead dev

Leadership

President June 2025 – May 2026
IVC Computer Science Student Organization
- Led a **9-member board** to execute member-driven initiatives, **partnering with a recruited tech lead** to secure hardware funding and **assist ∼10 students** during live technical workshops
- Instructed hands-on **Godot** workshops, mentoring members step-by-step from engine fundamentals to deploying their first playable web prototype on itch.io